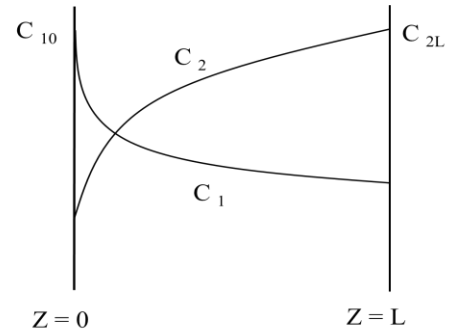


Practice Problems – Mass Transfer

- Imagine a film of stagnant liquid in which a chemical species of concentration C_2 is dissolved. At the surface $z = 0$, in contact with a solid wall, a concentration C_{10} of another species is maintained. The thickness of the film is L and it can be assumed that some external mechanism is present that permits us to maintain C_{10} at $z = 0$ and C_{2L} at $z = L$. Species 1 and 2 will diffuse and profiles $C_1(z)$ and $C_2(z)$ will be established at steady state.



Now further suppose that species 1 and 2 react at a rate $R_{12} = -k C_1 C_2$ (R_{12} is the rate of disappearance of 1 and 2). This reaction occurs homogeneously throughout the film. Species 1 and 2 diffuse through the film according to different diffusion coefficients, denoted by D_1 and D_2 respectively. Derive (do not solve) the governing equations (for one-dimensional diffusion only process) for the two species and state clearly the boundary conditions that reflect the physics of the process.

An analytical solution of these coupled nonlinear differential equations is not possible. However, one can make an approximation if it is assumed that the reaction rate is very fast i.e., when the diffusing molecules meet they react and disappear immediately. Let the plane at which they meet, $z = L_R$ is called the "reaction front". Derive analytical expressions for the concentration distributions of the two species for this case (in terms of L_R and other constants) within the respective domains of their existence and sketch the profiles.

Obtain an expression for the location of this "reaction front" i.e. an expression for L_R .

- Oxygen diffuses through the wall of drug containers and oxidizes many drugs rendering them inactive. It is the oxygen diffusion/reaction scenario that limits the shelf-life of many pharmaceutical products. To limit the oxidation of drugs, Oxygen scavengers, like sodium bisulfite (NaHSO_3) are often added to the drugs. The reaction to remove oxygen is:

$$2\text{HSO}_3^- + \text{O}_2 \rightarrow \text{SO}_4^{2-} + \text{H}_2\text{O}$$

Consider a liquid drug stored in a cylindrical polyethylene container. The container is 15 cm high and has an inner radius of 6 cm. The initial concentration of NaHSO_3 in the drug formulation is 1 g/lit. Neglect diffusion through the top and bottom of the container and assume that the NaHSO_3 reacts instantly with the O_2 . The O_2 concentration in air is also always 21%; and this may be approximated as the concentration at the outer end of the container. The effective diffusivity of O_2 through polyethylene is $9 \times 10^{-13} \text{ m}^2/\text{s}$ and you may assume $P = 1 \text{ atm}$, $T = 20^\circ\text{C}$, and the process operates at steady-state. Evaluate the concentration profile of O_2 in the container material. How thick must the walls of the container be to ensure that 90% of the NaHSO_3 remains after 1 year?

- A timed-release drug is dissolving in the intestine of a person. As a steady state approximation we may assume that the drug is a rod of overall radius r_o (m) and length L (m). The timed-release action is accomplished by putting an inert coating on the drug through which the drug diffuses with a diffusivity, D_{AB} . At the inner edge of the coating (r_i) the composition of the drug (mole fraction) is x_{ai} . On the outer surface, the digestive juices provide for mass transfer with a mass transfer coefficient of k_{ac} (m/s) (equivalent to convective heat transfer coefficient). The amount of drug within the intestine can be approximated as $x_{a\infty} \approx 0$. Assume that the drug is not released by

the two end-caps (circular) of the rod. You may also assume the total concentration of all species within the coating is c_t , a constant and it is a diffusion only process. Calculate the rate of drug release (mg/hr) in the intestine. Given: $r_i = 3\text{ mm}$, $r_o = 5\text{ mm}$, $D_{AB} = 10^{-10}\text{ m}^2/\text{s}$, $k_{ac} = 0.1\text{ m/s}$, $L = 5\text{ mm}$, $c_t = 0.4\text{ kg/m}^3$, $x_{ai} = 0.9$.

4. Two gases (O_2 and N_2) are separated into two bulbs connected by a narrow diameter capillary.

Pure N_2 then flows from right to left (with a velocity v), as shown in the figure. To what degree can O_2 diffuse upstream against the N_2 flow, i.e, derive the concentration distribution of O_2 in the tube. Assume that the system has achieved some steady state and that the volumes at the two ends are so large that each remains nearly pure, even though some exchange takes place (early stage steady state model). Find the solution for the limiting case of small vL/D , where D is the diffusion coefficient.

